

Binary search

Revise 360

Worksheet for the 360° experience: Binary search archive

Name: _____ Class: _____ Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Binary search archive. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Describe and trace binary search.
- Explain why sorted data is required.
- Compare binary and linear search choices.

Starter: before you put the headset on

In [2, 5, 8, 11, 14], the middle value is 8. Where can target 14 still be?

.....

.....

Key terminology

Binary search

.....

.....

Midpoint

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.....

The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Sorted data first

Turn right

Key facts in your own words:

Example to trace:

Sorted: 2, 5, 8, 11, 14

Middle value: 8

Target: 14

Keep the right half

Challenge: Why is sorted order required?

What must hold before binary search is used?

Binary search on unsorted data can...

2 Find the midpoint

Turn right

Key facts in your own words:

Example to trace:

low = 0; high = 5

mid = (low + high) DIV 2

mid = 2

Compare data[2]

Challenge: What is mid when low = 3 and high = 8?

low = 2, high = 6. Midpoint index?

low = 0, high = 6. Rounded-down midpoint?

3 Discard the wrong half

Turn around

Key facts in your own words:

Example to trace:

```
Target > data[mid]
low = mid + 1
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Target < data[mid]
high = mid - 1
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Challenge: Why use $\text{mid} + 1$ rather than mid ?

Match each comparison to the update.

Match each item to a description from the list below.

Report found / $\text{high} = \text{mid} - 1$ / $\text{low} = \text{mid} + 1$

Item	Matching description
Target greater	
Target smaller	
Target equal	

If $\text{target} < \text{data}[\text{mid}]$, which update is correct?

4 Trace a successful search

Turn around

Key facts in your own words:

Example to trace:

List: 3, 6, 9, 12, 15, 18, 21

Target: 18

mid 3: value 12

mid 5: value 18

Found in 2 comparisons

Challenge: What values are checked when searching for 6?

Search this list for 21. Checked values?

For [3,6,9,12,15,18,21], target 18 is at index...

5 Trace not found

Turn left

Key facts in your own words:

Example to trace:

List: 2, 4, 6, 8, 10

Target: 7

Checks: 6, 8

Then low = 3; high = 2

Result: not found

Challenge: Why does low > high mean not found?

For target 1 in [2, 4, 6, 8, 10], what is checked?

When low > high, the remaining range is...

6 Choose the search

Turn left

Key facts in your own words:

Example to trace:

Short unsorted list

One search: linear suits

Large sorted list

Many searches: binary suits

Challenge: Why not always sort before a one-off search?

Which is the strongest case for binary search?

Why can binary search scale well?

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Order the checks for target 14 in [2, 4, 6, 8, 10, 12, 14].

Put these steps in order: Check 14 at index 6 / Check 12 at index 5 / Keep indices 4 to 6 / Check 8 at index 3

What is the worst-case advantage as a sorted list grows?

Your score

Copy it from the score screen as soon as it appears.

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Key question

Why does a dictionary put words in order?

Review and improve

Use review mode. Explain one correction and why the correct answer fits.

Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Explain why binary search requires sorted data. [2]

b) Trace binary search for 4 in [2, 4, 6, 8, 10, 12, 14]. Round midpoint down. [3]

c) Explain when linear search could be a better choice. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Describe and trace binary search.			
Explain why sorted data is required.			
Compare binary and linear search choices.			